

Section 1: Introduction to Node.js

- What is Node.js and Why Use It?
- Installing Node.js (JS & TS setup)
- Running JavaScript outside the browser
- Using Core Modules (fs, path, http)
- File Operations (Sync & Async)
- Blocking vs Non-Blocking I/O
- Creating a simple Web Server (JS & TS)
- Routing Basics
- Building a Simple API
- HTML Templating (Static + Dynamic)
- Parsing URL Variables
- Custom Modules & NPM Basics
- Package Management, Versioning, Prettier + ESLint setup

Mini Project: *Basic HTTP server & API in Node.js with JS + TS*

Section 2: Web Fundamentals

- How the Web Works (Client-Server model)
- HTTP in Action
- Frontend vs Backend Development

- Static vs Dynamic Websites vs APIs
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Section 3: Node.js Architecture

- Node.js Internals (V8, Libuv, C++)
- Processes, Threads & Thread Pool
- Event Loop & Event-driven Architecture
- Streams (Theory + Practice)
- How `require` / `import` works (JS vs TS)

Mini Project: *Streaming large files efficiently*

Section 4: Asynchronous Programming

- Callback Hell & its Problems
- Promises (Creating & Consuming)
- Async/Await in Depth
- Returning Values & Handling Errors
- Working with Multiple Promises (Promise.all, race)

Mini Project: *File processing pipeline with Promises & Async/Await*

Section 5: Express.js Essentials

- Introduction to Express

- Setting up Express (JS & TS)
- REST API Design Principles
- CRUD Operations (GET, POST, PATCH, DELETE)
- Routing & Refactoring
- Middleware (Custom & 3rd-Party)
- Param Middleware & Chaining
- Serving Static Files
- Environment Variables & Config Management

Project: *User Management API with Express (JS & TS versions)*

Section 6: Databases with MongoDB

- Intro to MongoDB (Local + Atlas Cloud)
- CRUD Operations in MongoDB
- Using MongoDB Compass GUI
- Connecting Express with MongoDB
- Intro to Mongoose ORM
- Creating Models & Schemas
- CRUD with Mongoose

Mini Project: *Tours API – Create, Read, Update, Delete*

Section 7: Advanced API Features

- MVC Architecture in Node.js
- Filtering, Sorting, Pagination
- Aliasing & Refactoring Query Features
- Aggregation Pipelines
- Virtual Properties & Middleware in Mongoose
- Data Validation (Built-in & Custom)

Project: *Scalable Tours API with Advanced Query Features*

Section 8: Error Handling & Debugging

- Debugging Node.js with ndb
 - Handling Unhandled Routes
 - Global Error Handling Middleware
 - Errors in Async Functions
 - 404 & Production vs Development Errors
 - Database-specific Error Handling (Validation, Duplicates)
 - Handling Unhandled Rejections & Uncaught Exceptions
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Section 9: Authentication & Security

- User Modeling & Password Management
- JWT Authentication (Sign Up, Login, Protecting Routes)

- Authorization (Role-based Access Control)
- Password Reset Functionality (Email + Tokens)
- Nodemailer + SendGrid Email Integration
- Security Best Practices:
 - Rate Limiting
 - Secure HTTP Headers
 - Data Sanitization
 - Preventing Parameter Pollution
 - JWT Cookies & CSRF considerations

Project: *Auth System with JWT, Roles, and Secure Password Resets*

Section 10: Advanced Data Modeling

- MongoDB Data Modeling Strategies
- Geospatial Data (Locations, Maps)
- Embedding vs Referencing Documents
- Populating & Virtual Populating
- Nested Routes in Express
- Factory Functions for CRUD Handlers
- Improving Performance with Indexes
- Aggregation for Ratings & Stats
- Geospatial Queries

Project: *Tours + Reviews API with Nested Routes & Advanced Modeling*

Section 11: Server-Side Rendering

- Server-side vs Client-side Rendering
 - Using Pug Template Engine
 - Base Layouts, Blocks, and Partial
 - Building Tour Pages & User Dashboard
 - Integrating Mapbox (Geospatial Visualization)
 - Auth with Views (Login, Logout, Protecting Pages)
 - Error Pages & User Account Management
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Section 12: File Uploads & Payments

- Image Uploads with Multer
- Resizing & Processing Images (Sharp)
- Multiple Image Uploads for Tours
- Complex Email Handling (Pug + Nodemailer)
- Payment Integration with Stripe
- Bookings Model & API
- Handling Checkout Success & Stripe Webhooks

Project: *Full Booking System with Payments & Email Notifications*

Section 13: Deployment & DevOps

- Git & GitHub Workflow
 - Preparing for Production (Config, Env Vars)
 - Deploying Node.js Apps (Heroku, Render, etc.)
 - HTTPS Setup & Security Testing
 - Handling Process Signals (SIGTERM, SIGINT)
 - CORS & Cross-Origin Setup
 - Production Monitoring & Logs
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Section 14: TypeScript with Node.js

- Setting up TypeScript in a Node.js Project
- tsconfig & Compiler Options
- Using Types for Express, MongoDB & Middleware
- Creating Type-safe Models & DTOs
- Error Handling with Type Safety
- Migrating a JS Express API → TS

Mini Project: *TypeScript-based Node.js REST API*

Section 15: Final Capstone Project

- Building a **Production-ready “Natours” Clone (Tours Booking App)**

- Features:
 - Authentication + Authorization
 - Tours + Reviews API
 - Payments with Stripe
 - Email Notifications
 - File Uploads & Image Processing
 - Geospatial Queries + Maps
 - Secure Deployment
 - Deploy final app on **Heroku / AWS / Render**
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👉 By the end, learners will:

- Master **Node.js Internals & Asynchronous Programming**
- Build **RESTful APIs with Express (JS & TS)**
- Use **MongoDB + Mongoose** for data modeling
- Implement **Authentication, Authorization, Security**
- Work with **SSR (Pug), File Uploads, Payments**
- Deploy full-stack apps with **best practices**